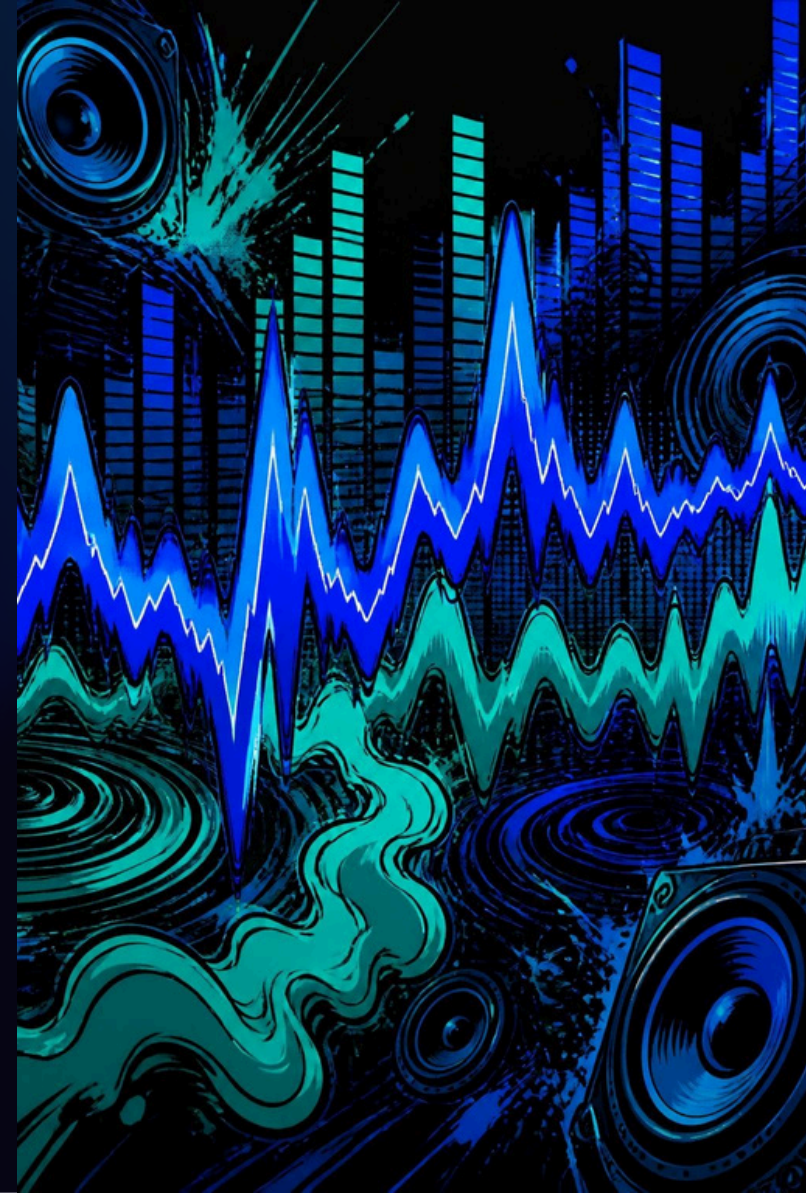


Spotify Tracks Analytics

Data Visualization Project — exploring artists, genres, popularity, and audio features through interactive visualizations.

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Project Overview

This project analyzes the Spotify Tracks Dataset using **Python, Pandas, Plotly, and Streamlit** to uncover insights into music trends, artist popularity, and audio characteristics.



Clean

Prepare and validate the raw dataset



Explore

Perform Exploratory Data Analysis (EDA)



Visualize

Build interactive Plotly charts



Deploy

Develop a Streamlit dashboard

Dataset at a Glance

Spotify Tracks Dataset

Sourced from **Kaggle**, this dataset provides a rich foundation for music analytics across genres and audio dimensions.

114,000 Songs

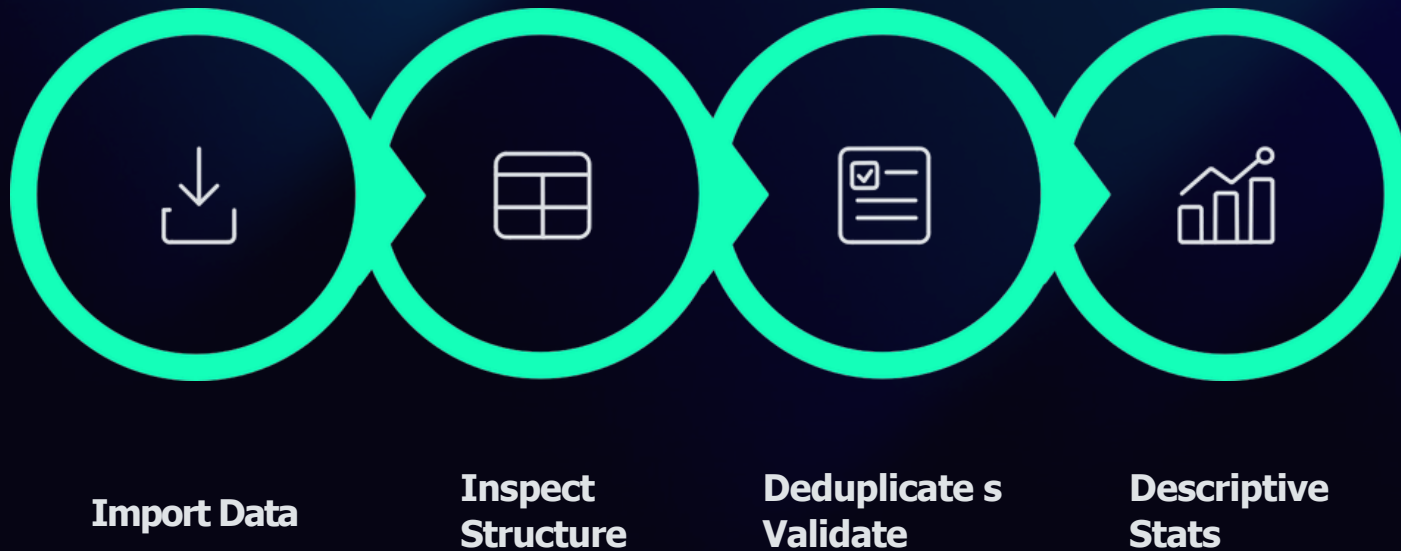
Spanning multiple music genres

20 Features

Artist, track, album, popularity, danceability, energy, tempo, valence, acousticness, explicit content

Data Cleaning s Preprocessing

A rigorous preprocessing pipeline ensured the dataset was clean, consistent, and ready for analysis.



📌 The outcome: a clean, reliable dataset with no missing values, duplicates removed, and all data types validated.

Exploratory Data Analysis

Five core analyses reveal patterns in artist popularity, genre distribution, and audio feature relationships.

→ **Top 10 Artists by Average Popularity**

→ **Genre Popularity Distribution**

→ **Average Popularity by Musical Key**

→ **Danceability vs Energy**

→ **Explicit vs Non-Explicit Songs**

Deeper Visual Insights

Additional visualizations explore song structure, genre diversity, and feature correlations across the full dataset.

**Song
Duration
Distribution**

**Songs by
Genre
(Treemap)**

**Audio
Feature
Correlation
Heatmap**

**Musical Key
Distribution**

Genre → Artist Sunburst Chart

Key Findings

Artist Popularity

Certain artists consistently achieve higher average popularity scores across the dataset.

Genre Variation

Popularity varies significantly across music genres, reflecting listener preferences.

Danceability & Energy

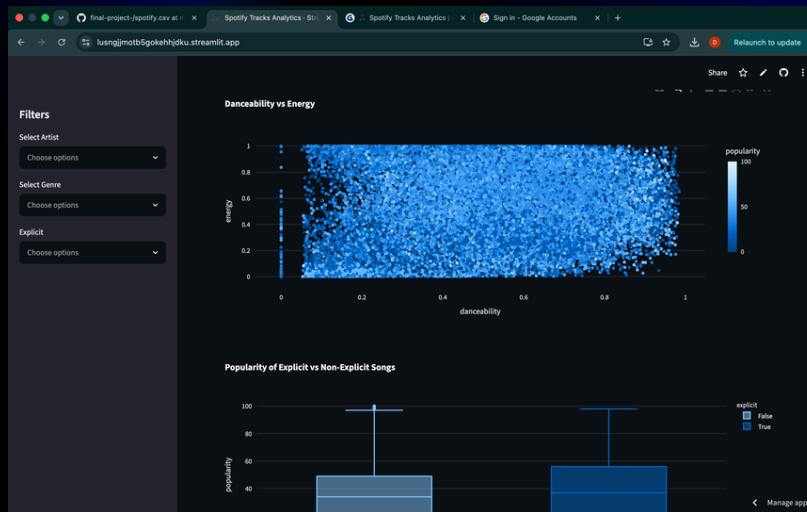
A positive correlation exists between danceability and energy — upbeat tracks tend to be more energetic.

Explicit Content

Explicit and non-explicit songs exhibit distinct popularity distributions.



Streamlit Interactive Dashboard



Dashboard Features

The Streamlit app enables dynamic, user-driven exploration of the full dataset.

Sidebar Filters

Filter by Artist, Genre, and Explicit Content

KPI Cards

Total Songs, Artists, Genres, and Average Popularity

10 Plotly Charts

Fully interactive visualizations responding to user filters

Conclusion

This project demonstrates the full data science workflow — from raw data to interactive insights — using industry-standard Python tools.

01

Analyzed the Spotify Tracks Dataset

114,000 songs across 20 audio features cleaned and explored

02

Built Interactive Visualizations

Plotly charts revealing trends in artists, genres, and audio features

03

Deployed a Streamlit Dashboard

An accessible, filterable interface for real-time data exploration

04

Demonstrated Python in Practice

Applied Pandas, Plotly, and Streamlit in a cohesive end-to-end project



Thank You!

Questions and feedback are welcome.

Presented by
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